
CanaryRAG: Detection, Attribution, and Active Defense for Retrieval-Augmented Generation Systems

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Abstract

Retrieval-Augmented Generation (RAG) is widely used to ground large language models in external document collections, but it introduces new security risks at the retrieval layer. Prior work has shown that RAG systems are vulnerable to attacks such as document poisoning, probing queries, and membership inference. Existing defenses often rely on filtering inputs or post-processing outputs, which typically occur after sensitive documents have already been retrieved. We propose CanaryRAG, a system-level defense that embeds synthetic canary documents into a RAG knowledge base and treats their retrieval behavior as a security signal. We extend canary-based detection with multi-dimensional attribution and an active honeypot based defense that injects controlled honey-text and tracks adversarial persistence. This approach enables earlier detection, interpretable attribution, and lightweight mitigation with minimal impact on benign usage.

1 Problem Statement

Retrieval-Augmented Generation enhances language models by retrieving external documents prior to generation. While effective, this architecture expands the attack surface of deployed systems. Recent work demonstrates that attackers can exploit retrieval through document poisoning, probing queries, and membership inference attacks that influence which documents are retrieved rather than manipulating model outputs directly.

Despite growing interest in RAG security, most defenses operate at the output level or offline. These approaches provide limited insight into adversarial intent and often intervene after sensitive content has already been retrieved. This motivates runtime defenses that operate directly at retrieval time and expose interpretable security signals.

2 Motivation and Relevance

We observe that retrieval behavior itself can indicate abnormal system interaction. Inspired by canary tokens and honeypots in traditional security, CanaryRAG embeds synthetic canary documents into the retrieval corpus. These documents are designed to be irrelevant to benign queries and should not appear under normal usage.

Retrieval of a canary document therefore signals unusual or adversarial interaction. This system-level approach aligns with the goals of adversarial and trustworthy foundation models by emphasizing monitoring, interpretability, and mitigation without modifying model weights or inspecting user content.

3 Technical Approach

The project builds sequentially from detection to attribution and mitigation within a standard RAG pipeline.

3.1 Baseline RAG and Canary Insertion

We first implement a baseline RAG system using an off-the-shelf embedding model, a vector database (e.g., FAISS or Chroma), and an open source language model. During the insertion phase, we embed multiple types of synthetic canary documents to enable attribution:

- **Keyword-dense canaries** to detect document poisoning or retrieval boosting.
- **Semantically general canaries** to detect broad probing or membership inference.

3.2 Retrieval-Time Detection and Attribution

The retriever is instrumented to log retrieval time signals such as canary presence in the top-k results, rank, similarity scores, and frequency over time. These signals are used to trigger detection rules and provide non-binary attribution.

When a canary is retrieved, a lightweight verifier model (the LLM-Gatekeeper) evaluates whether the query canary relationship is logically consistent or indicative of adversarial behavior. This enables differentiation between attack types without altering the core retrieval pipeline.

3.3 Active Defense via Honeypot Strategy

If detection and attribution indicate suspicious behavior, the system applies an active defense. Instead of blocking the query, CanaryRAG injects controlled honey-text believable but false information into the generation context.

Subsequent queries from the same session are monitored to measure adversarial persistence. If follow up queries continue to reference the injected honey text, this provides a high confidence signal of malicious intent, reducing false positives while extending the system from detection to mitigation.

4 Expected Outcomes

We expect canary documents to be rarely retrieved under benign usage but to surface more frequently under adversarial interaction. Multi-dimensional attribution is expected to clarify the nature of detected attacks, while adversarial persistence should distinguish malicious users from accidental triggers. Overall, the project aims to demonstrate that combining retrieval time signals with attribution and honeypot based mitigation enables earlier and more informative RAG security defenses.

5 Timeline and Feasibility

Weeks 1–2 will focus on building the baseline RAG system and inserting canary documents.

Weeks 3–4 will involve retrieval instrumentation and baseline data collection.

Weeks 5–6 will focus on attribution logic, the LLM-Gatekeeper, and attack simulation.

Weeks 7–8 will be used for evaluation and report writing.

The project is feasible within a semester and does not require training large models, proprietary data, or access to model internals. All components rely on existing open source tools, and the scope is intentionally focused on system level behavior.

References

Relevant references will include prior work on Retrieval-Augmented Generation, RAG security attacks, and canary or honeypot based detection methods.